MILITARY SPECIFICATION

FUZE, IMPACT, PYROTECHNIC, M733; DELAY ASSEMBLY FOR

LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of delay assembly, for use on Fuze, Impact, Pyrotechnic, M733.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1234- Pyrotechnics Sampling, Inspection and Testing

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9323256 - Delay Assembly

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Material, - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on Drawing (Dwg.) 9323256, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. - All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection.

4.3.1 Submission. - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Delay Assembly	9323256	200

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63136 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Delay Assembly	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Delay Assembly (Dwg 9323256)				
	Examination for defects	25			4.4.2.1
	Static test:				
*	Burn time at hot temperature	25		3.2	4.5.2
	Burn time at ambient temperature	25		3.2	4.5.2
*	Burn time at cold temperature	25		3.2	4.5.2
	Static smoke emission (projectiles)	50		3.2	4.5.3
	Smoke emission (cartridges)	50		3.2	4.5.4

NOTES: * Burning time at hot and cold phases shall be recorded for informational purposes.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition each lot shall contain the following:

a. Delay composition from one batch from one manufacturer.

b. Output composition from one batch from one manufacturer.

c. Ignition composition from one batch from one manufacturer.

d. Parts from one interfix lot number from one manufacturer.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63136 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Delay Assembly		9323256
			NEXT HIGHER ASSEMBLY 9323255

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth to ignition composition		0.40%	3.2	Gage
102	Static tests:				
	(a) Burn time at hot temperature			3.2	4.4.3.2.1, 4.4.3.3.1, 4.5.2
	(b) Burn time at ambient temperature			3.2	4.4.3.2.2, 4.4.3.3.2, 4.5.2
	(c) Burn time at cold temperature			3.2	4.4.3.2.3, 4.4.3.3.3, 4.5.2
103	Static smoke emission (projectiles)			3.2	4.4.3.4, 4.4.3.5, 4.5.3
104	Smoke emission (cartridges)			3.2	4.4.3.6, 4.4.3.7, 4.5.4
Minor					
201	Total length		0.65%	3.2	Gage

NOTES:

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4.4.3 Testing.

4.4.3.1 Moisture content (See Table I). -

Material	Drawing
Delay composition	9323270
Ignition composition	9323269
Output composition	9323271

At the time and place of loading the contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification at least one (1) sample of each material from each eight hour's production of cartridges. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all cartridges loaded with the non-forming material shall be rejected.

4.4.3.2 Burn time of delay assembly. - Beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have been accepted, sixty (60) samples shall be selected from each lot and tested as follows:

4.4.3.2.1 Hot temperature. - Twenty (20) samples shall be subjected to this test.

4.4.3.2.2 Ambient temperature. - Twenty (20) samples shall be subjected to this test.

4.4.3.2.3 Cold temperature. - Twenty (20) samples shall be subjected to this test.

4.4.3.2.4 Rejection. - If three (3) or more samples in ambient phase fail the applicable requirement, the lot represented by the sample shall be rejected. Burning time in hot and cold phase shall be recorded for informational purposes.

4.4.3.3 Burn time of delay assembly, (Regular Production). - After three (3) acceptable lots have met the criteria of 4.4.3.2, the Government inspector shall select twenty-four (24) samples from each lot and tested as follows:

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4.4.3.3.1 Hot temperature. - Eight (8) samples shall be subjected to this test.

4.4.3.3.2 Ambient temperature. - Eight (8) samples shall be subjected to this test.

4.4.3.3.3 Cold temperature. - Eight (8) samples shall be subjected to this test.

4.4.3.3.4 Rejection. - If two (2) or more samples in ambient phase fail the applicable requirement, the lot represented by the sample shall be rejected. Burning time in hot and cold phase shall be recorded for informational purposes.

4.4.3.4 Static smoke emission of delay assembly (Projectiles). - Beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have been accepted, thirty-two (32) assemblies shall be selected from each lot, assemble into fuzes and then assemble into projectiles and subjected to this test. If one (1) or more assemblies fail to comply with the drawing requirement, the lot shall be rejected.

4.4.3.5 Static smoke emission of delay assembly (Regular Production) (projectiles). - After three (3) acceptable lots have met the criteria of 4.4.3.4, the Government inspector shall select thirteen (13) assemblies from each lot, assemble into fuzes and then assemble into projectiles and subjected to this test. If one (1) or more assemblies fail to comply with the drawing requirement, the lot shall be rejected.

4.4.3.6 Smoke emission of delay assembly (cartridges). - Beginning with the first lot produced and continuing until thru (3) acceptable lots (lots not accepted by disposition) have been accepted, thirty-two (32) assemblies shall be selected from each lot, assemble into fuzes and then assemble into cartridges and subjected to this test. If one (1) or more assemblies fail to comply with the drawing requirement, the lot shall be rejected.

4.4.3.7 Smoke emission of delay assembly, Regular Production, (cartridges). - After three (3) acceptable lots have met the criteria of 4.4.3.6, the Government inspector shall select thirteen (13) assemblies from each lot, assemble into fuzes, assemble into cartridges and subjected to this test. If one (1) or more assemblies fail to comply with the drawing requirement, the lot shall be rejected.

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4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5.1 Moisture content.

4.5.1.1 Delay composition, ignition composition and output composition. - The moisture content of each of these compositions at the time and place of loading shall be determined in accordance with Method 102.1 of MIL-STD-1234 or any other approved method.

4.5.2 Static burn time of delay assembly hot, ambient and cold temperature phases. - The delay assembly shall be temperature conditioned at each phase for 2 hours and static tested using a Government approved procedure.

4.5.3 Static smoke emission of delay assembly (projectiles). - The delay assembly shall be assembled into fuzes and then into projectiles and static tested using a Government approved procedure. Observation shall be made for drawing requirement.

4.5.4 Smoke emission of delay assembly (cartridges). - The delay assembly shall be assembled into a fuze and then into a cartridge. The cartridge shall be fired from an approved launcher, placed on an approved mount to impact at 300 meters minimum from the muzzle of the launcher on a six inch thick loose gravel pad. The gravel used, shall have a 1/2 inch maximum size. Observation shall be made for the drawing requirement.

5. PACKAGING

5.1 Preservation and packaging. - The delay assembly shall be preserved and packaged in accordance with an approved moisture proof container.

5.2 Marking. - Marking of the containers shall be in accordance with Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Shipping. - When components from more than one lot are shipped as a carload, each lot shall be kept separate, and

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the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - The component covered by this specification is intended for use on Fuze, Impact Pyrotechnic, M733.

6.2 Ordering Data. - See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. - See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRDAR-QAA-I, ARRADCOM, Dover, NJ 07801.

6.4 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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